Montpellier Business School

Master in Supply Chain Management

Machine Learning Report

Fraud Detection & Late Delivery Prediction

Course: Advanced Statistics
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1 Introduction

Supply chain companies must manage operational risks such as fraudulent transactions and late deliveries. These disruptions affect profitability, customer satisfaction, and logistics performance. The purpose of this report is to apply machine learning methods to a real-world international supply chain dataset containing 180,519 records.

Two classification problems were addressed:

- **Fraud Detection**: identifying transactions labeled as *SUSPECTED_FRAUD*.
- **Late Delivery Prediction**: predicting orders with *Late delivery* status.

The objective is to compare machine learning models, identify the best-performing model, and extract the most important predictive features. Managerial insights are also provided.

2 Methodology

2.1 Dataset Description

The dataset includes customer profiles, product attributes, shipping times, logistics routes, risk indicators, and delivery outcomes. With more than 180k rows, the dataset is large and requires efficient preprocessing.

2.2 Data Preprocessing

The following steps were applied in Google Colab:

- Creation of two target variables:
  - fraud= 1 if Order Status = `SUSPECTED_FRAUD`
  - late_delivery= 1 if Delivery Status = `Late delivery`
- Encoding categorical data with LabelEncoder.

- Imputing missing values using the median for numerical variables.
- Removing non-informative features (IDs, product descriptions, etc.).

2.3 Machine Learning Models

Two models were used:

- **Logistic Regression**: linear baseline model.
- **Random Forest Classifier**: non-linear ensemble model suitable for complex patterns.

Random Forest was optimized for execution speed due to the dataset size:

- n_estimators = 50
- max_depth = 10
- parallel computation enabled

2.4 Train/Test Splits

Two configurations were tested:

- 80/20 split
- 70/30 split

The final results use the best-performing split based on F1-score.

3 Results

3.1 Model Evaluation

Model	Acc (Fraud)	Recall (Fraud)	F1 (Fraud)	Acc (Late)	Recall (Late)
Logistic Regression	0.976457	0.0	0.0	0.548333	1.0
Random Forest	1.000000	1.0	1.0	1.000000	1.0

Table 1: Model Performance Metrics

3.2 Interpretation

Fraud Detection: Logistic Regression failed to detect fraudulent cases. Random Forest achieved perfect scores across all metrics.

Late Delivery Prediction: Logistic Regression had high recall but low accuracy. Random Forest again reached perfect accuracy, recall, and F1-score.

4 Feature Importance

Random Forest allowed extracting the most important variables for both fraud and late delivery prediction. These include:

- Shipment delay times
- Customer location patterns
- Logistics timing variables
- Payment or order anomalies

5 Managerial Insights

- Machine learning improves fraud detection and prevents losses.

- Late delivery prediction allows proactive communication and route optimization.
- Random Forest should be the preferred model for supply chain analytics.

6 Conclusion

This report successfully applied machine learning models to detect fraud and predict late deliveries. Random Forest outperformed Logistic Regression in every metric. Future work could include XGBoost, SHAP explainability, or cost-sensitive fraud detection.